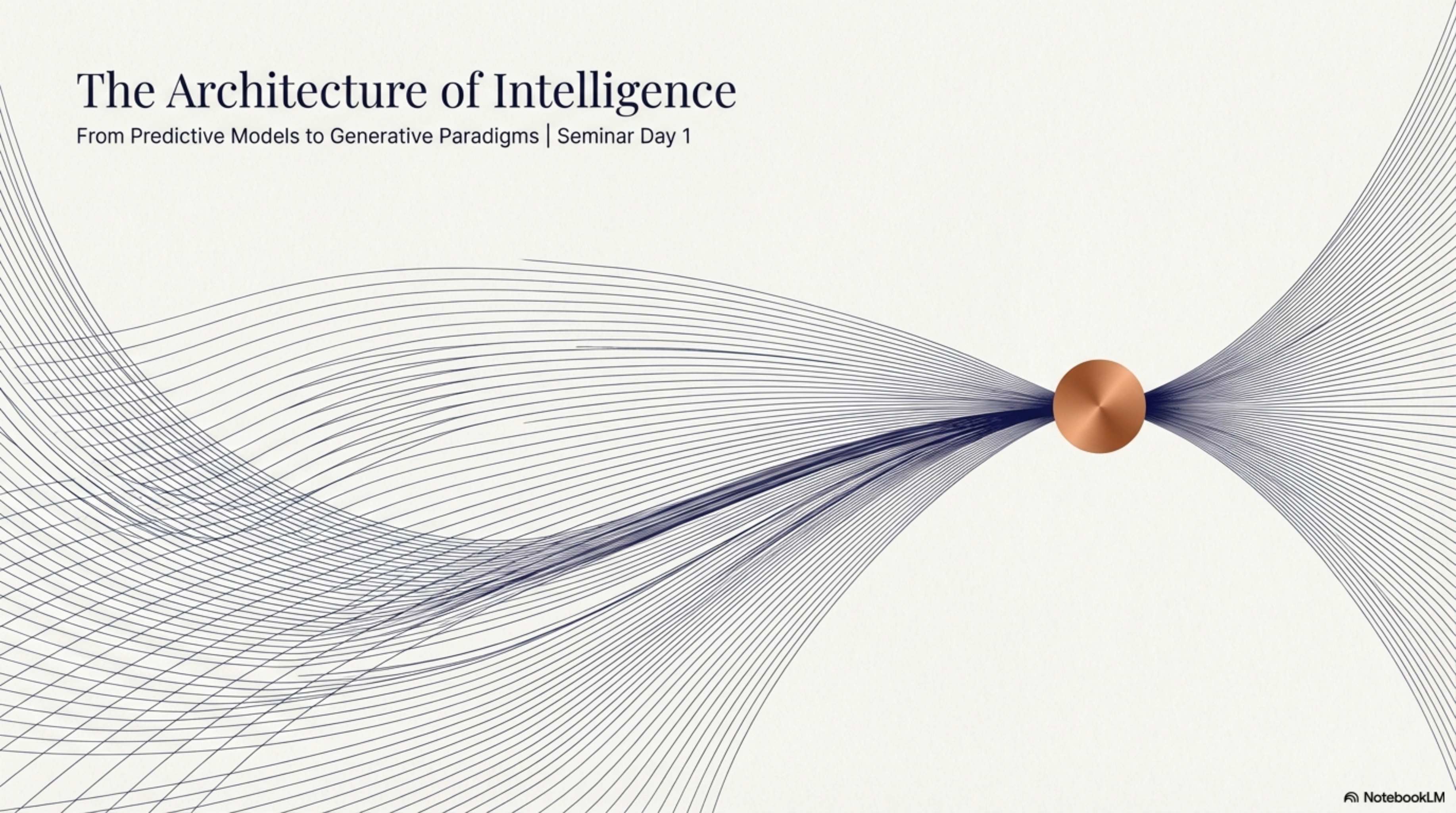
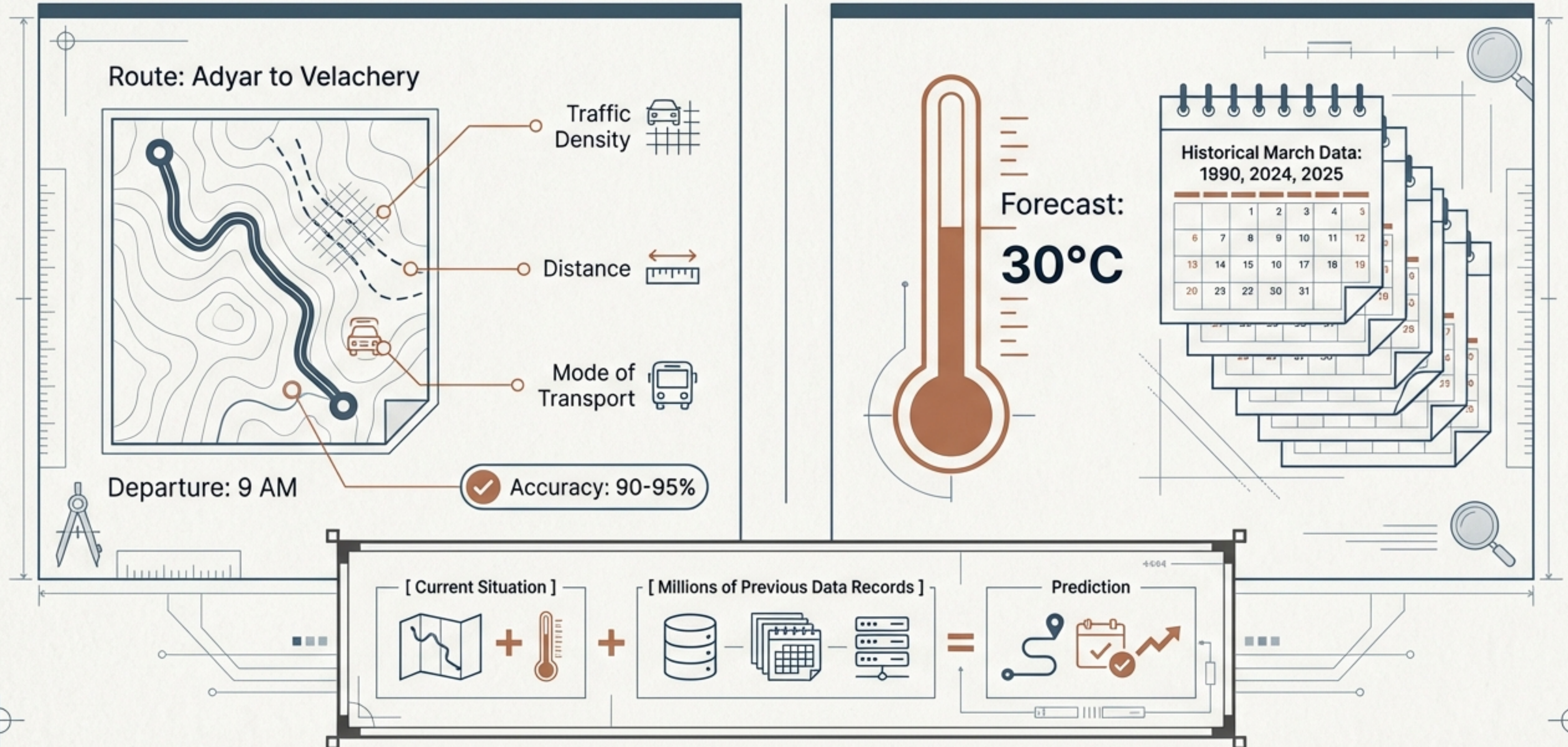


The Architecture of Intelligence

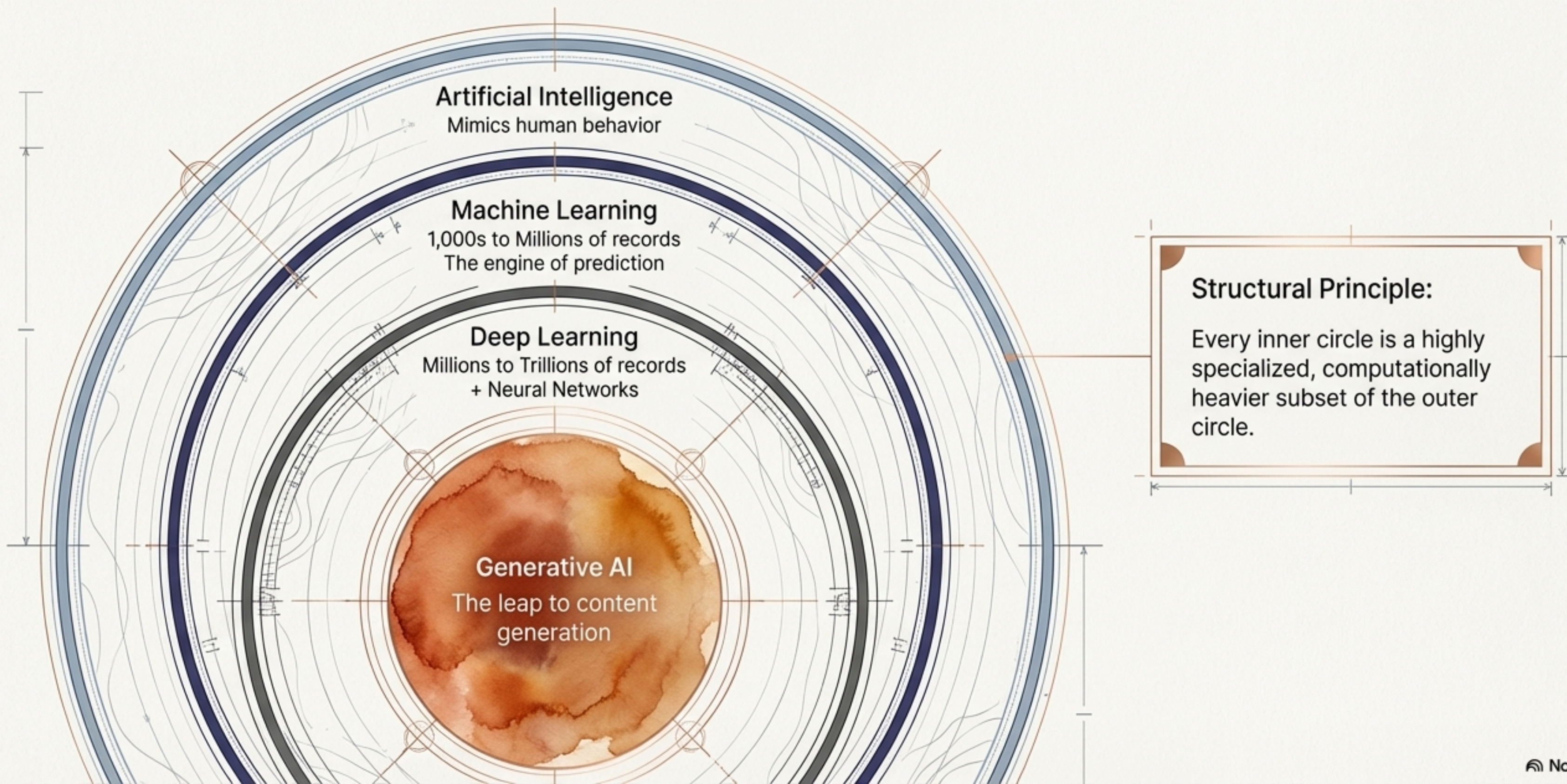
From Predictive Models to Generative Paradigms | Seminar Day 1



Mimicking Human Behavior to Navigate the Unknown

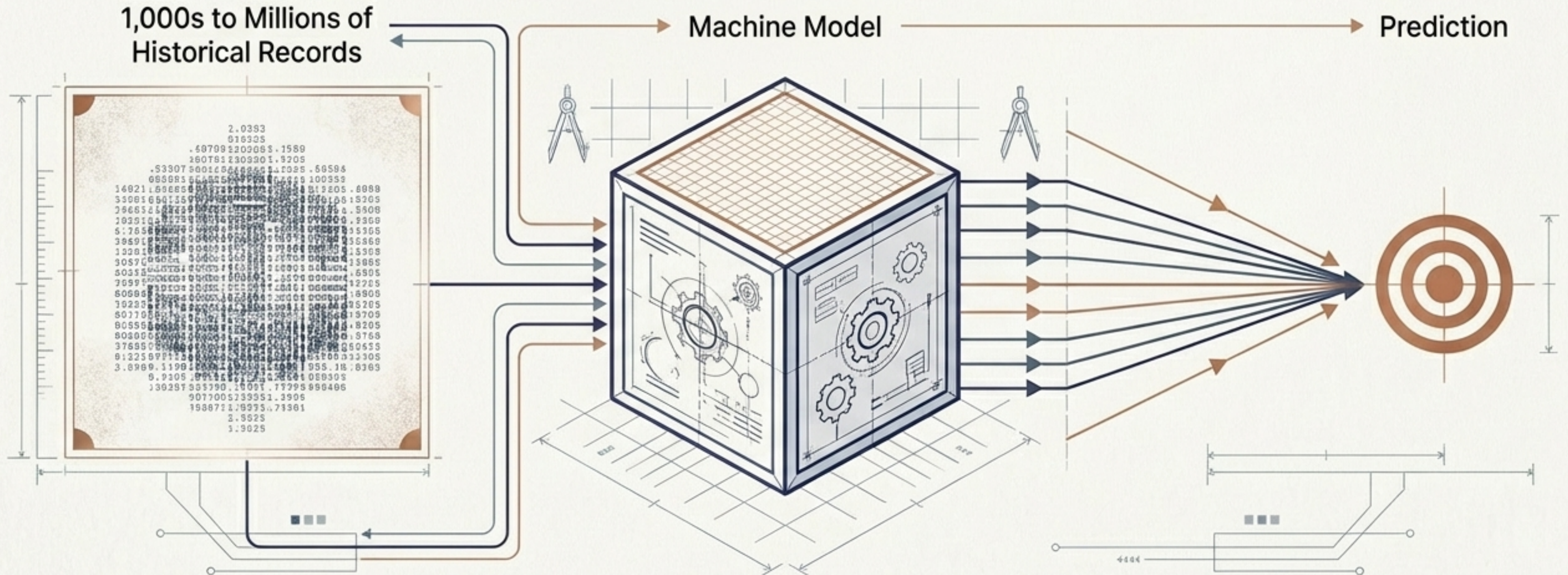


The Landscape of Artificial Intelligence

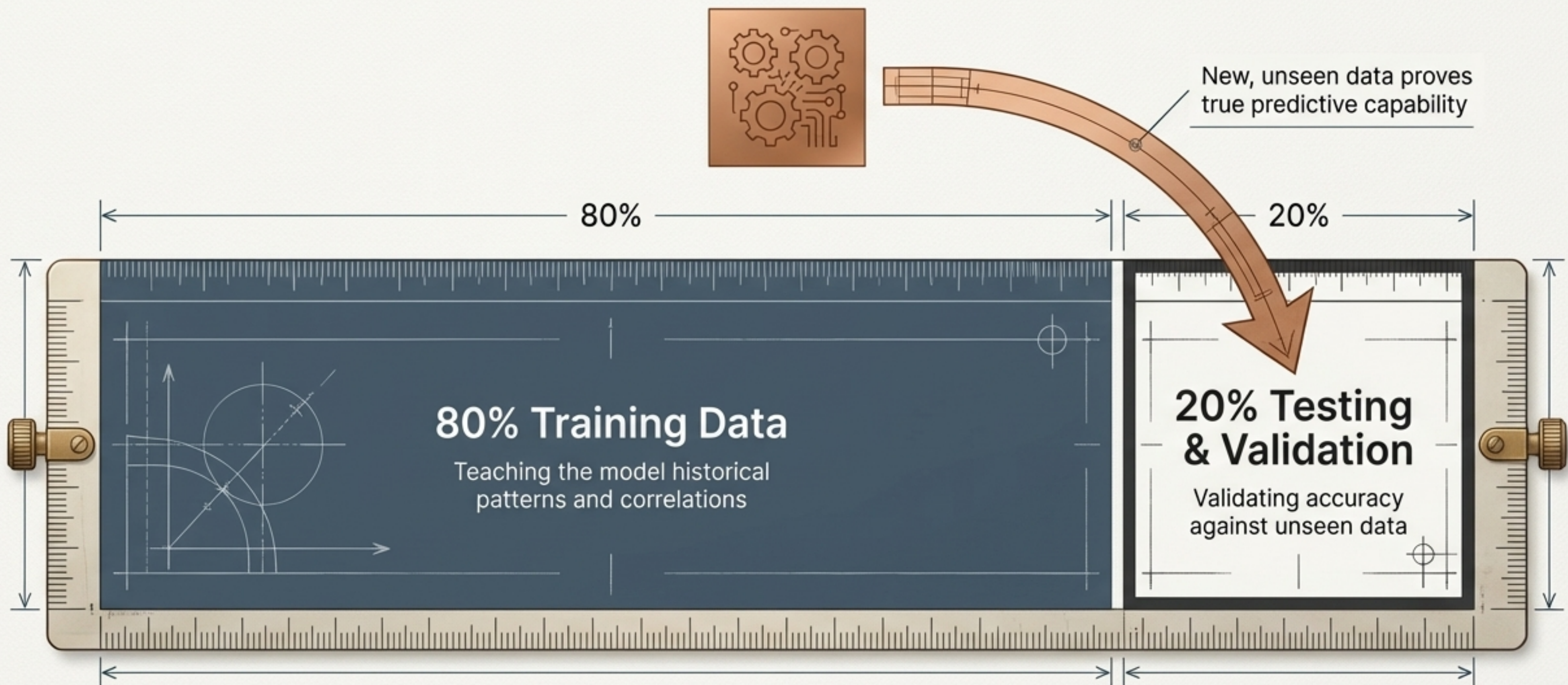


Machine Learning: The Engine of Prediction

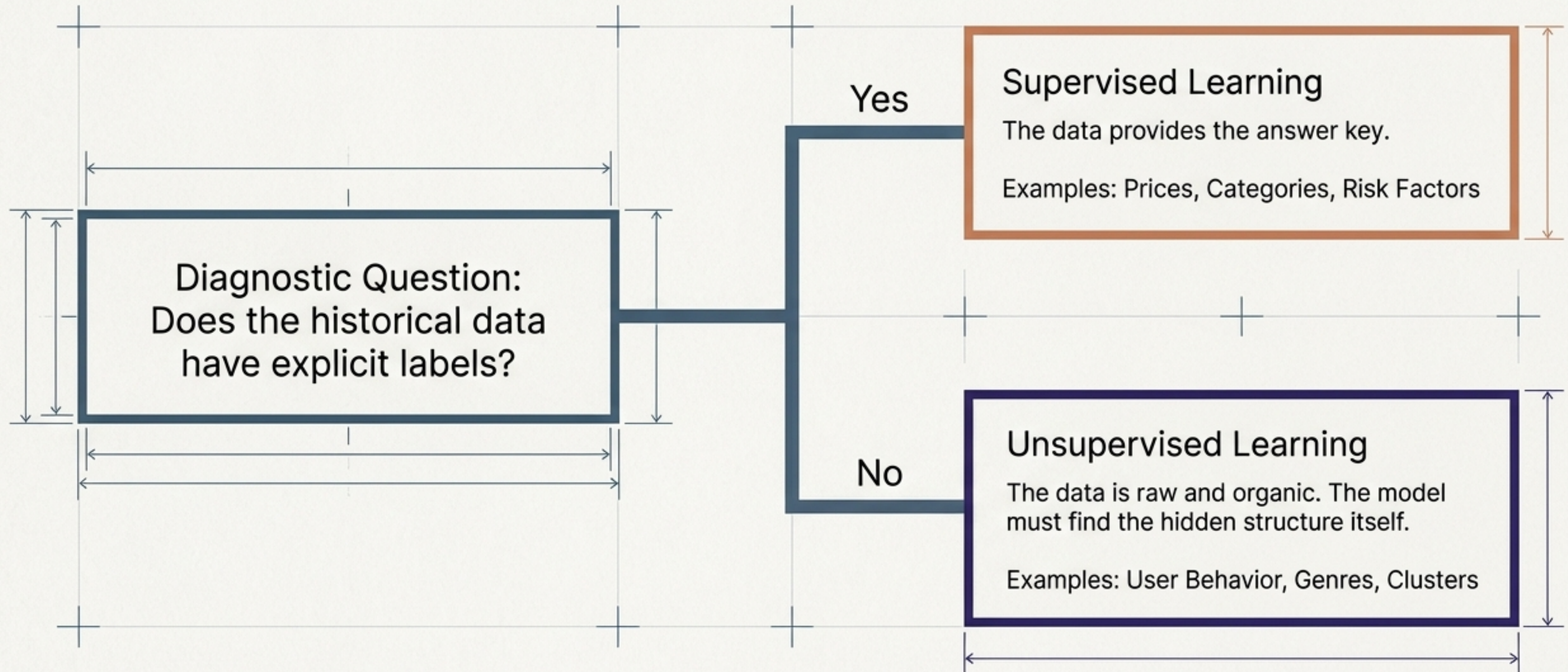
Machine Learning is the subset of AI that autonomously digests historical datasets to identify patterns and predict future outcomes.



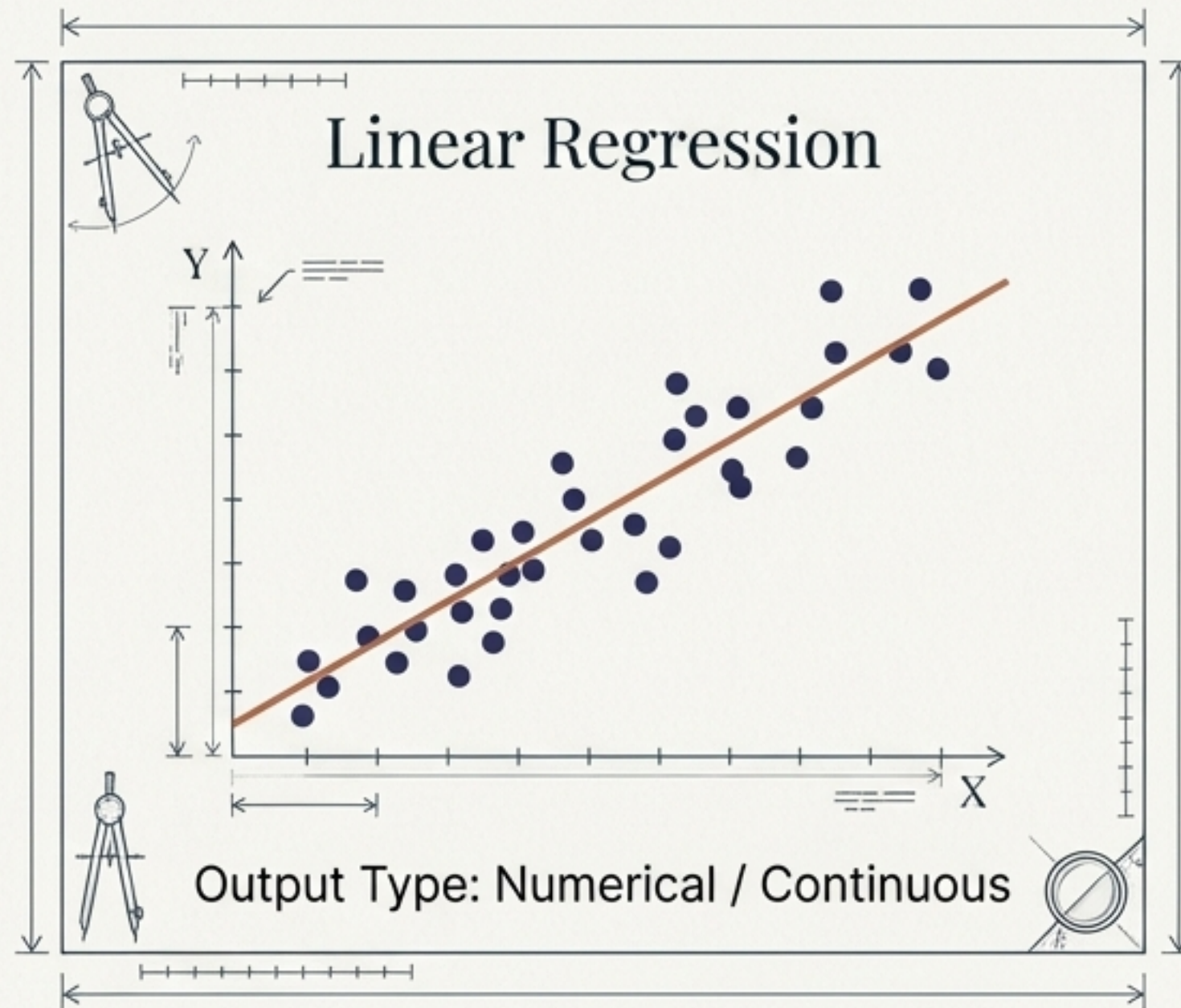
Building Predictive Power: The 80/20 Rule



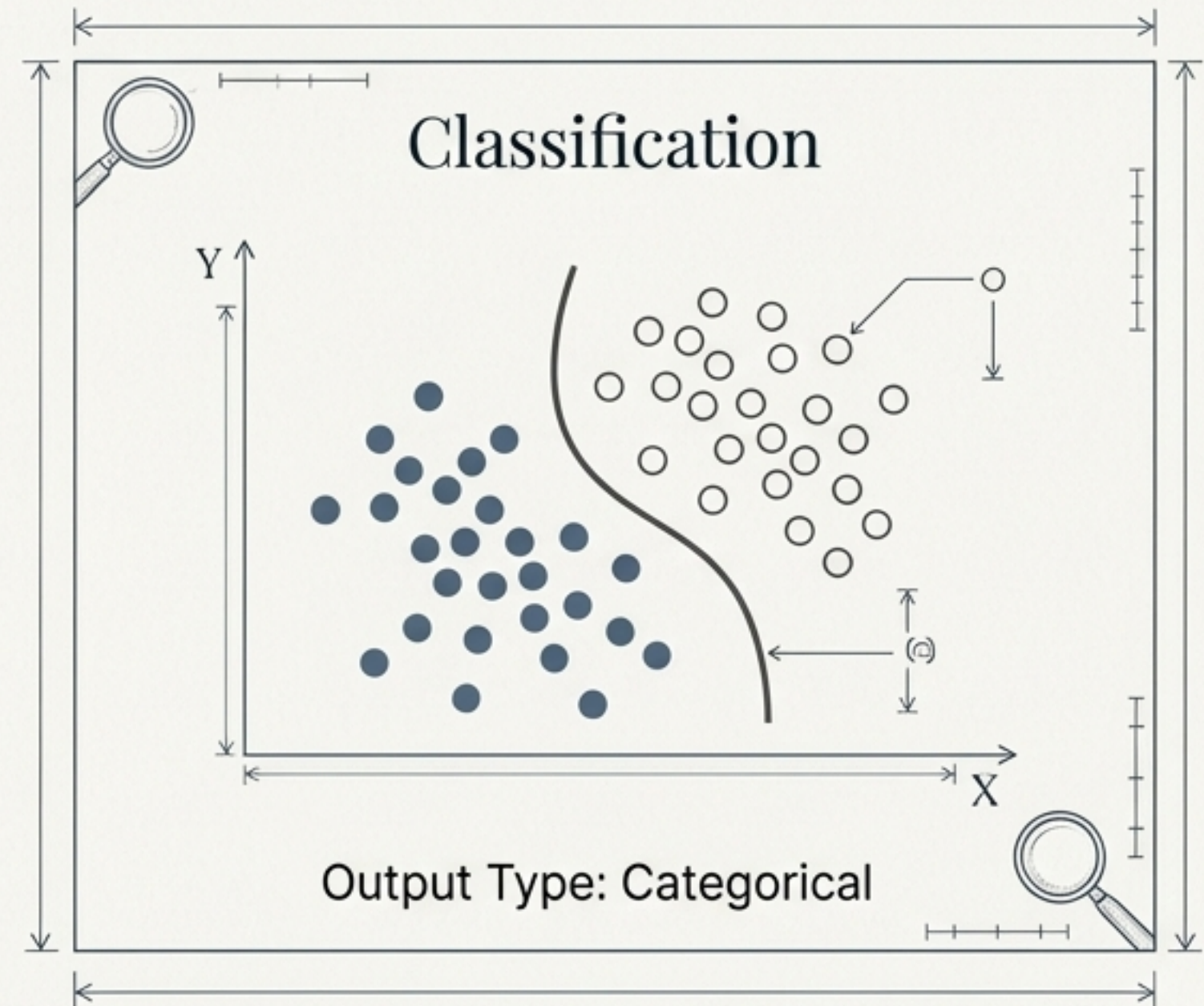
The Machine Learning Playbook in Playfair Display



Supervised Learning: Predicting with Precision



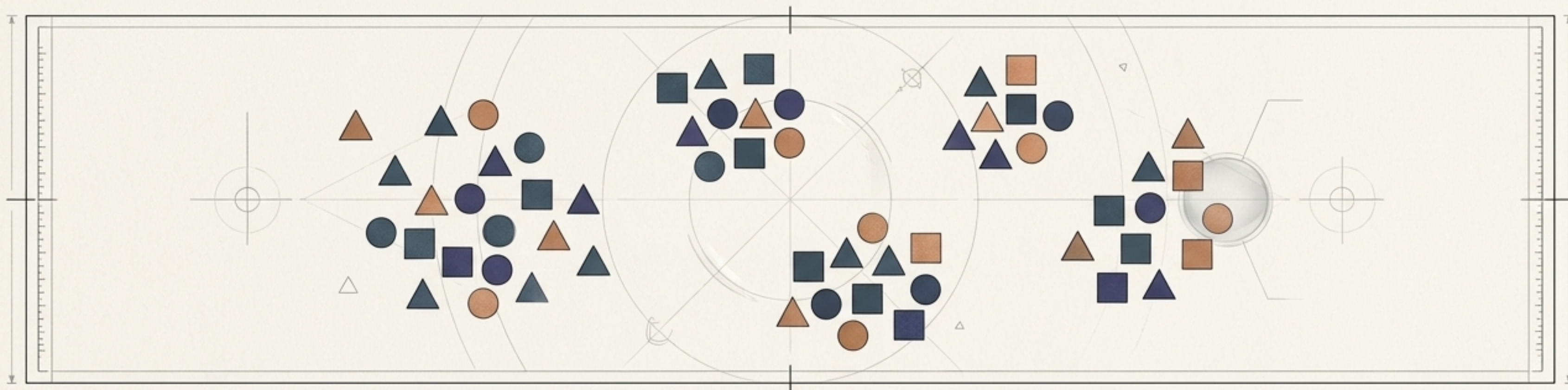
- Home Purchase / Real Estate Pricing
- Numerical Defect Prediction



- Heart Attack Risk Prediction (Yes/No)
- Categorical Defect Classification

Unsupervised Learning: Discovering Hidden Structures

Processing unlabelled data to group behaviors and genres organically.



Collaborative Filtering

Type: Unsupervised

Mechanism: Driven by organic, unlabelled behavior without explicit user input.

Examples: Watch history, Time spent

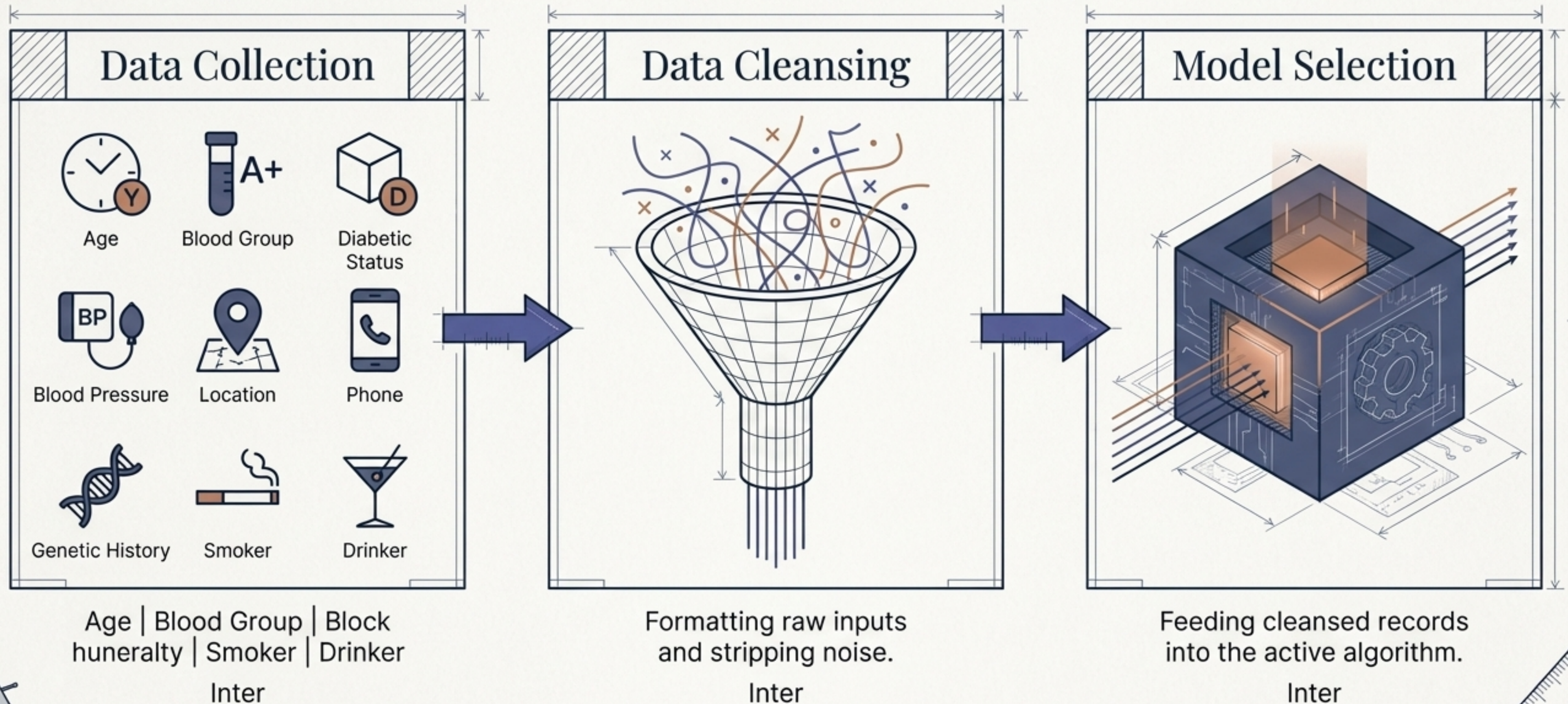
Content-Based Filtering

Type: Supervised

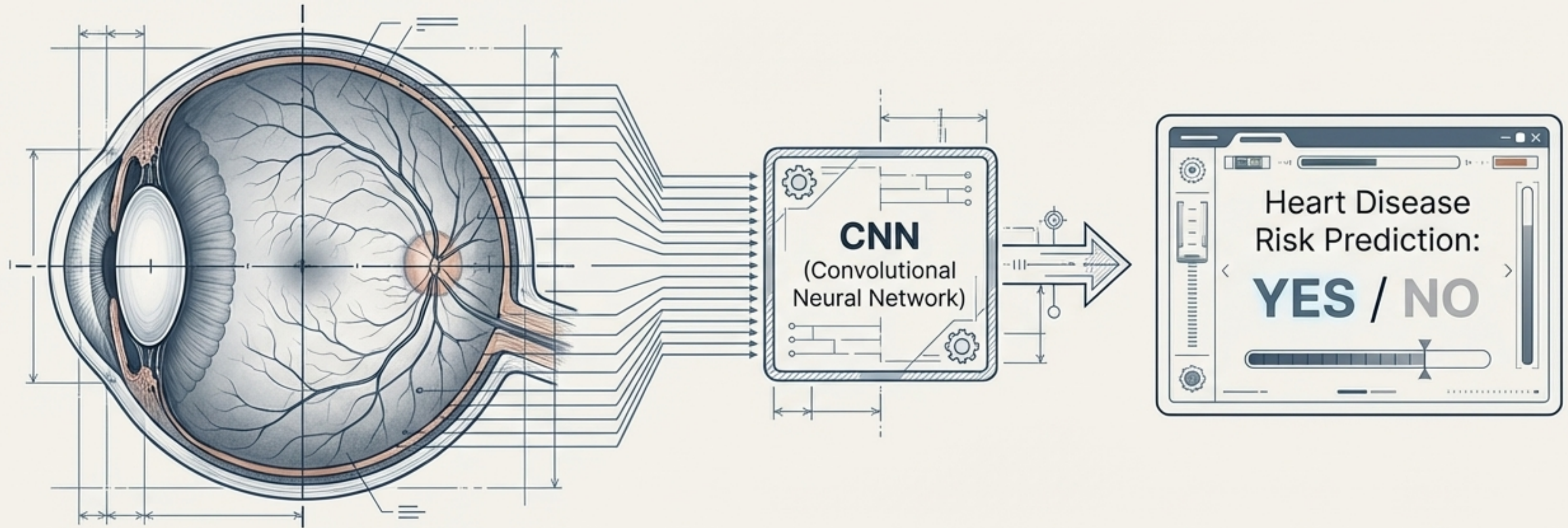
Mechanism: Relies on explicit labels provided by the user.

Examples: Likes, Dislikes, Star Ratings

The Real-World Pipeline: Heart Disease Prediction



Advancing to Multimodal Inputs: Retinal Scanning



Multimodal AI

Expanding inputs beyond text and numbers to accept images and video.

The CNN

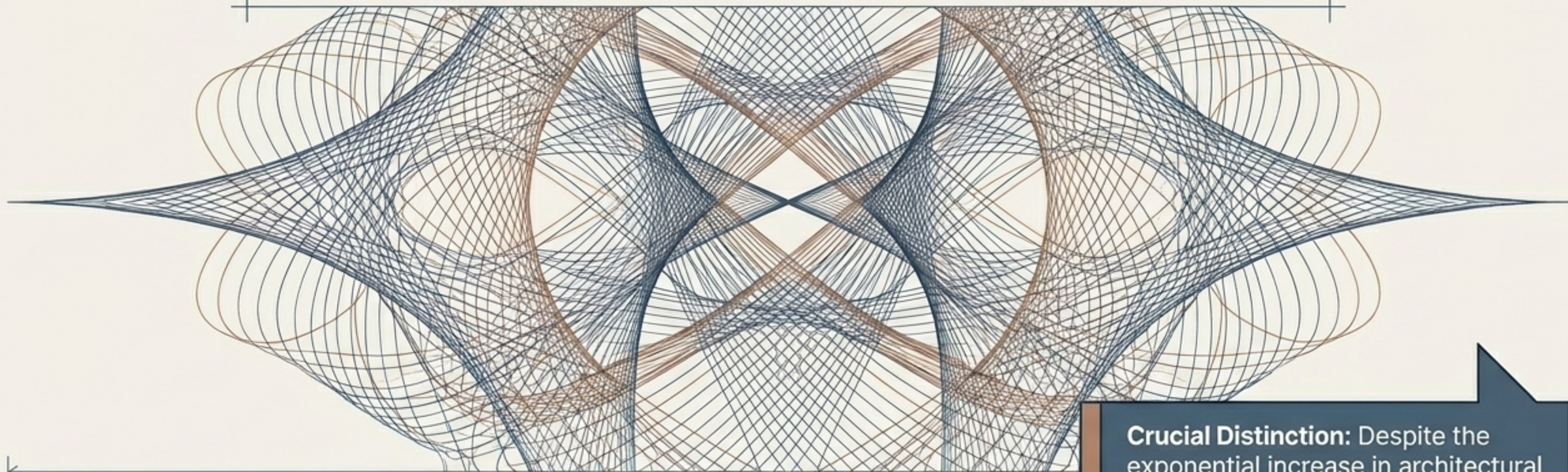
Specifically selected to read and process millions of historical retinal images.

The Output

A completely new, unlabelled retinal image instantly generates a risk prediction.

Scaling to Deep Learning

Scale: Millions to Trillions of Records



$$\left[\begin{array}{c} \text{Machine} \\ \text{Learning} \end{array} \right] + \left[\begin{array}{c} \text{Vast} \\ \text{Data} \end{array} \right] + \left[\begin{array}{c} \text{Neural} \\ \text{Networks} \end{array} \right] = \text{Deep Learning}$$

Crucial Distinction: Despite the exponential increase in architectural complexity and data scale, Deep Learning at this stage is still exclusively predictive. It analyzes the known to forecast the unknown.

The Paradigm Shift: From Prediction to Generation

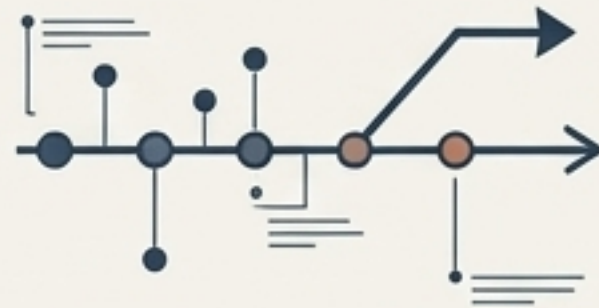
Predictive AI

1. The Metaphor:



The Lens
(Converging Focus)

2. The Goal:



Analyze history to
predict the known

3. The Output:

	✓	✓	YES
✓	✗	✗	\$
✓	✗	\$	VALUES

Numbers and Categories
(Yes/No, \$ Values)

4. The Technology:



Machine Learning &
Predictive Deep Learning

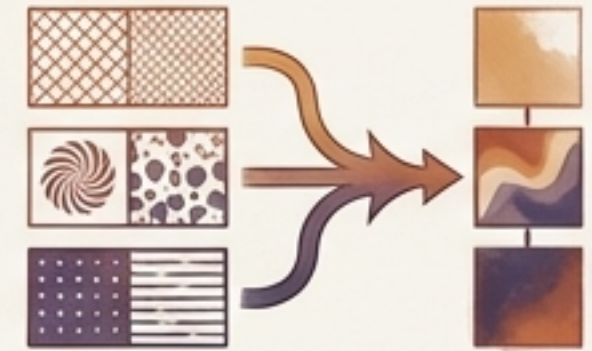
Generative AI

1. The Metaphor:



The Canvas
(Expansive Emergence)

2. The Goal:



Synthesize patterns to
generate the entirely new

3. The Output:



Net-new Content
(Images, Text, Audio, Video)

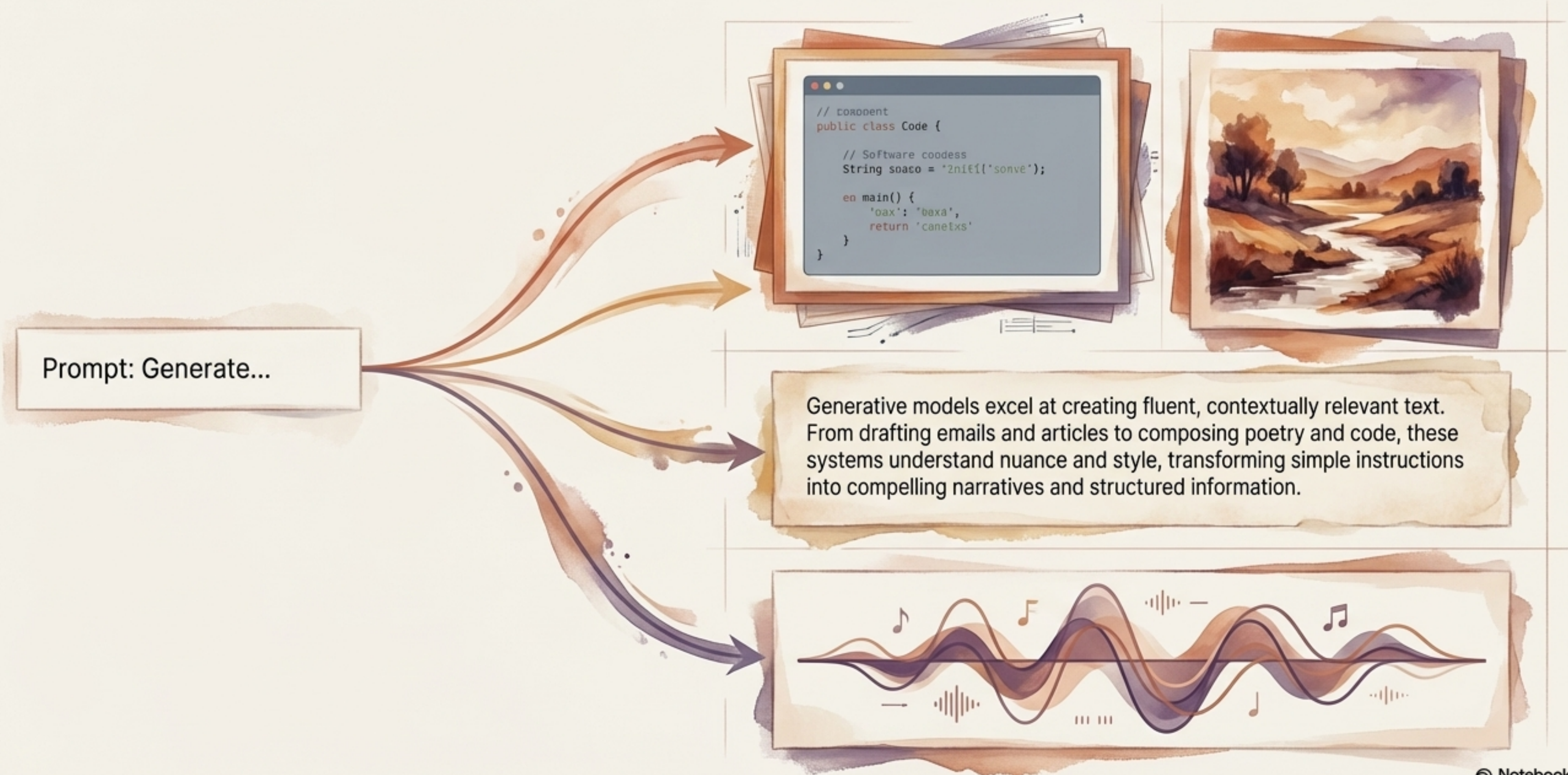
4. The Technology:



A specialized, non-predictive
subset of Deep Learning

Demystifying Generative AI

Generative AI operates across all domains. It is not limited to text; it builds the content of the future from the ground up.



The Engine of Generation: GPT Architecture



The Framework

GPT (Generative Pretrained Transformer)
Pretrained on colossal datasets spanning all fields,
not just software.

The Provider

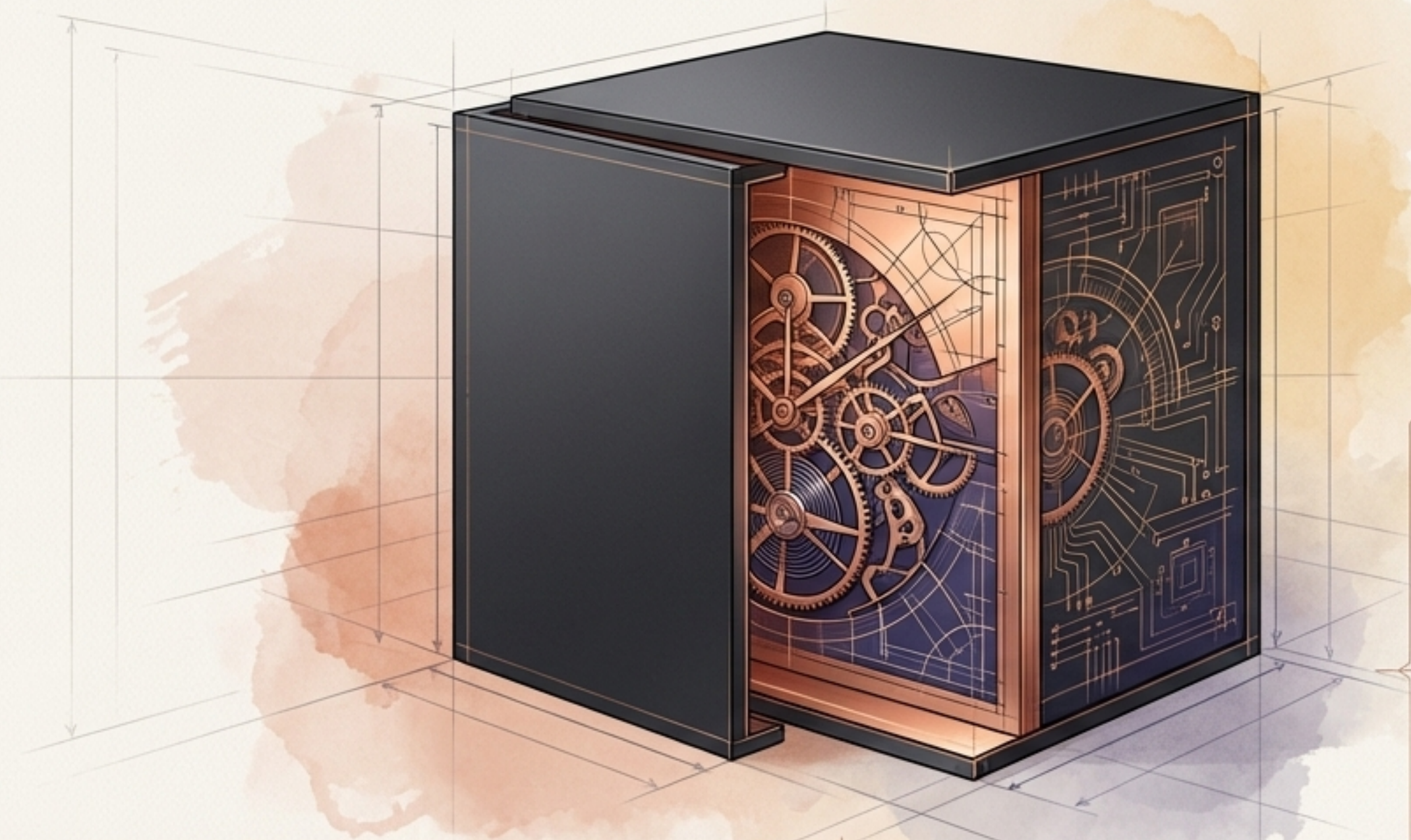
Driven by OpenAI
Primary Chat Agent: ChatGPT

The Scale of Evolution

GPT-4: 96 to 120 Internal Layers
GPT-5: 190 Layers powering 120 Billion Parameters

Beyond the Black Box

in Playfair Display



Coming Up on Day 2:

LLM Anatomy and Internal Workings in Playfair Display

Tomorrow, we open the black box. We will deconstruct the internal anatomy of Large Language Models (LLMs) and map exactly how generative text is constructed layer by layer.

Journey: Evolution → Prediction → Generation